

A logarithmic lower bound for Nyström estimates

Siavash Sadeghi

4 October 2026

Abstract

A fixed smooth amplitude on a smooth closed curve with a straight arc gives growth of order square root of $\log k$. This addresses the arbitrary smooth amplitudes and general parametrizations allowed in the target; it does not assert a counterexample for every physical boundary-integral kernel.

Provenance and review. Separated and polished from the supplied AI-assisted research note prepared with ChatGPT. Codex reviewed the mathematical argument and accompanying checks. This is a research claim awaiting independent mathematical review; no independent human audit or formal proof verification is claimed.

1 Entry 523: the square-root logarithm is necessary

We use exactly the general-amplitude formulation in catalogue entry 523 [1]. The original paper and its fourth version retain the general-curve conjecture [2]. It is enough to disprove the estimate at $s = 0$.

For clarity, the target uses

$$L_{1,k}(t, \tau) = k \int_{S^1} e^{ik\langle \gamma(t) - \gamma(\tau), \omega \rangle} f_k(\omega, t, \tau) dS(\omega), \quad \widehat{L}_{1,k,m}(t) = \frac{1}{\sqrt{2\pi}} \int_0^{2\pi} e^{-im\tau} L_{1,k}(t, \tau) d\tau.$$

All amplitude derivatives must be uniformly bounded in k . It suffices to disprove the asserted bound at $s = 0$, where the target quantity is $k^{-1} \sum_{0 < |m| \leq (1-\varepsilon)N} \|\widehat{L}_{1,k,m}\|_{L^2(\Gamma)}$. The construction uses a single smooth amplitude independent of k and the admissible choices $\varepsilon = 1/2$ and $N = 4\lceil k \rceil$.

1.1 A fixed smooth curve and a fixed amplitude

Choose a smooth embedded closed curve containing a straight horizontal arc, with a smooth periodic parametrization satisfying, on $|t| < 1/4$,

$$\gamma(t) = (\sigma(t), 0), \quad \sigma(t) = t + \frac{1}{2}t^2. \quad (1)$$

Such a curve is obtained by smoothly completing a line segment away from this chart and then making a positive smooth reparametrization. The local speed is $1 + t > 0$; the global speed is smooth, bounded above, and bounded away from zero. No positive-curvature or unit-speed hypothesis is imposed by the repository target.

Let χ be a smooth periodic cutoff supported in $(-1/4, 1/4)$ and equal to one on $[-1/8, 1/8]$. Parametrize a neighbourhood of $(-1, 0) \in S^1$ by

$$\omega(\theta) = (-\cos \theta, \sin \theta).$$

Choose a nonnegative even smooth cutoff ψ , supported in $|\theta| < \theta_0$ for a sufficiently small fixed $\theta_0 > 0$, and equal to one near zero. Define, independently of k ,

$$f_k(\omega(\theta), t, \tau) = \psi(\theta)\chi(t)\chi(\tau), \quad (2)$$

and extend by zero on the rest of the circle. This is a fixed smooth amplitude; every derivative required in the question is uniformly bounded in k .

Put $\rho = m/k$ and restrict it to

$$J = \left[1 - \frac{1}{32}, 1 + \frac{1}{32}\right].$$

For θ_0 small enough, $\tau_\theta = \rho / \cos \theta - 1$ lies in the region where $\chi(\tau) = 1$, uniformly for $\rho \in J$.

1.2 Reduction to a nearly degenerate one-dimensional integral

Using the local coordinate and interchanging the two compact integrals,

$$\widehat{L}_{1,k,m}(t) = \frac{k\chi(t)}{\sqrt{2\pi}} \int \psi(\theta) e^{-ik\sigma(t)\cos\theta} \int \chi(\tau) e^{ik[\sigma(\tau)\cos\theta - \rho\tau]} d\tau d\theta. \quad (3)$$

The inner phase is exactly quadratic in τ , has critical point τ_θ , and has second derivative $\cos \theta > 0$. Since the cutoff is identically one near the critical point, quadratic stationary phase, with integration by parts away from that point, gives for every $B > 0$ a remainder $O(k^{-B})$, uniformly in the parameters. With $u = 1 + t$, the result is

$$\widehat{L}_{1,k,m}(t) = \chi(t)\sqrt{k} e^{i\pi/4} \int \frac{\psi(\theta)}{\sqrt{\cos\theta}} e^{ik\Psi(u,\rho,\theta)} d\theta + O(k^{-B}), \quad (4)$$

where

$$\Psi(u, \rho, \theta) = \rho - \frac{1}{2}u^2 \cos \theta - \frac{\rho^2}{2 \cos \theta}. \quad (5)$$

All constants here concern the one fixed curve and amplitude.

Choose a fixed small $\delta_0 > 0$, for example $1/32$, and take

$$t = \rho - 1 - \delta, \quad k^{-1/4} \leq \delta \leq \delta_0.$$

These t lie in the plateau of χ . Also $u = \rho - \delta < \rho$. The angular phase has just one stationary point on the support, at zero, and

$$\Psi_{\theta\theta}(u, \rho, 0) = -A, \quad A = \frac{\rho^2 - u^2}{2} = \frac{\delta(2\rho - \delta)}{2} \asymp \delta.$$

The degeneration $A \downarrow 0$ as $t \uparrow \rho - 1$ is responsible for the logarithm.

1.3 Uniform stationary phase on the required range

For completeness, we justify uniformity as A tends to zero with k . Let

$$g(\theta) = \Psi(u, \rho, 0) - \Psi(u, \rho, \theta), \quad b(\theta) = \frac{\psi(\theta)}{\sqrt{\cos\theta}}.$$

For $|\theta| < \theta_0 < \pi/2$,

$$g''(\theta) = \frac{\rho^2}{2 \cos \theta} - \frac{u^2 \cos \theta}{2} + \frac{\rho^2 \sin^2 \theta}{\cos^3 \theta} \geq A > 0. \quad (6)$$

Further,

$$g(\theta) = \frac{A\theta^2}{2} + O(\theta^4),$$

with the remainder uniform in u, ρ in the present compact ranges. Rescale $\zeta = \sqrt{kA} \theta$. On each bounded ζ interval,

$$kg\left(\frac{\zeta}{\sqrt{kA}}\right) = \frac{\zeta^2}{2} + O\left(\frac{\zeta^4}{kA^2}\right), \quad b\left(\frac{\zeta}{\sqrt{kA}}\right) \rightarrow 1$$

uniformly, because $kA^2 \gtrsim k^{1/2} \rightarrow \infty$.

The tails of this rescaled integral are also uniform. On the positive half-line, the derivative of the rescaled phase is increasing and is at least ζ by (6); the negative half-line is analogous. The amplitudes have uniformly bounded total variation. Integration by parts therefore bounds either tail $|\zeta| > R$ by C/R , uniformly in k, A, u, ρ . First letting $k \rightarrow \infty$ on a bounded interval and then $R \rightarrow \infty$ proves

$$\int b(\theta) e^{ik\Psi(u, \rho, \theta)} d\theta = \frac{e^{ik\Psi(u, \rho, 0)}}{\sqrt{kA}} \left(\sqrt{2\pi} e^{-i\pi/4} + o(1) \right) \quad (7)$$

uniformly for $\rho \in J$ and $k^{-1/4} \leq \delta \leq \delta_0$.

Combining (4) and (7) gives

$$\widehat{L}_{1,k,m}(\rho - 1 - \delta) = \sqrt{\frac{2\pi}{A}} e^{ik\Psi(u, \rho, 0)} (1 + o(1)). \quad (8)$$

In particular, for all sufficiently large k ,

$$\left| \widehat{L}_{1,k,m}(\rho - 1 - \delta) \right|^2 \geq \frac{c_1}{\delta}$$

with $c_1 > 0$ uniform in the specified ranges.

1.4 The logarithmic lower bound

On this arc, arclength measure is $ds = (1+t) dt = u dt$, with u bounded above and below by positive constants. Thus, uniformly for integers m with $m/k \in J$,

$$\left\| \widehat{L}_{1,k,m} \right\|_{L^2(\Gamma)}^2 \geq c_2 \int_{k^{-1/4}}^{\delta_0} \frac{d\delta}{\delta} \geq c_3 \log k \quad (9)$$

for sufficiently large k . There are at least $c_4 k$ such positive integers m .

Choose $\varepsilon = 1/2$ and $N = 4\lceil k \rceil$. The prescribed sum includes all these integers, and at $s = 0$ its extra weight is one. Hence

$$\frac{1}{k} \sum_{0 < |m| \leq (1-\varepsilon)N} \left\| \widehat{L}_{1,k,m} \right\|_{H_k^0(\Gamma)} \geq c_5 \sqrt{\log k} \rightarrow \infty. \quad (10)$$

Theorem 1.1. *The constant bound in the repository's general smooth-amplitude formulation of entry 523 is false. A fixed smooth curve, a fixed smooth nonconstant-speed parametrization, and a fixed smooth amplitude already force the square-root logarithmic loss.*

Remark 1.2. The amplitude localization in (2) is explicitly permitted by the repository statement. This argument does not identify that amplitude with every particular physical single- or double-layer kernel. It also does not contradict the positive-curvature, unit-speed result, since the construction has a flat arc and variable speed.

References

- [1] M. Colbrook, AIM catalogue, [entry 523: Removing the logarithmic loss in Helmholtz Nyström estimates](#). Repository statement checked 4 October 2026.
- [2] J. Galkowski, M. Rachh, and E. A. Spence, *Helmholtz boundary integral methods and the pollution effect*, [arXiv:2507.22797v4](#) (2026), equations (9.3)–(9.6), Lemmas 11.1 and 11.3, and Conjecture 11.4.